INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

MEATBALLS, BEEF AND RICE, IN SPICY TOMATO SAUCE, SMOKE FLAVORING ADDED, THERMOSTABILIZED, FOR MEAL, READY-TO-EAT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers thermostabilized meatballs, beef and rice, in spicy tomato sauce, smoke flavoring added in flexible pouches for use as a component of the Meal, Ready-to-Eat, Individual.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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SPECIFICATIONS

MILITARY

MIL-P-44073 - Packaging and Thermoprocessing of Foods in Flexible Pouches

STANDARDS

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Institutional Meat Purchase Specifications for Fresh Beef, Series 100

(Copies are available from the Chief, Livestock and Meat Standardization Branch, Livestock and Seed Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2603, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

Meat and Poultry Inspection Regulations

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

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U.S. Standards for Grades of Tomato Paste

(Copies are available from the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 0709, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. Standards for Milled Rice

(Copies are available from the Federal Grain Inspection Service, Animal and Plant Health Inspection Service (APHIS), Printing, Distribution, and Mail Section, Room G-110, Federal Building, 68005 Belcrest Road, Hyattsville, MD 20782.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, 650 California Street, Suite 800, San Francisco, CA 94108.)

AOAC International

Official Methods of Analysis of AOAC International

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(Application for copies should be addressed to AOAC International, 2200 Wilson Boulevard, Suite 400-CD, Arlington, VA 22201-3301.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-flavors, off-odors, and off-colors.

* 3.2.1 Beef. The beef shall be derived from any one or any combination of the following USDA Certified Institutional Meat Purchase Specifications (IMPS) cuts: Item Number 114 (chuck, shoulder clod), 115 (chuck, square-cut, boneless), 128 (chuck, cross-cut, boneless), 133 (triangle, boneless), 161 (round, shank-off, boneless), 163 (round, shank-off, 3-way, boneless), 165 (round, rump and shank-off, boneless), 167A (round, knuckle, peeled), 169 (top round), 170A (bottom round, heel-out), and 171B (outside round). All beef shipped between plants shall be accompanied by a USDA, Agricultural Marketing Service (AMS) Certificate to certify condition of the product and compliance with the IMPS Quality Assurance Provisions.

* 3.2.1.1 Handling and storage. Handling and storage of the boned and trimmed beef, prior to processing into the finished product, shall be in accordance with the following requirements:

a. Beef processed on the day of initial certification shall be maintained in the temperature range of 28°F to 50°F (inclusive).

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b. Holding in the fresh-chilled state for not more than 4 days after initial certification is permitted provided the beef is maintained in the temperature range of 28°F to 40°F (inclusive).

c. Holding in the frozen state for not more than 180 days after initial certification is permitted provided that the beef is:

-frozen to 0°F or lower within 72 hours after initial certification
-stored at 0°F or lower
-protected from freezer deterioration and damage
-stored in containers that are adequate to maintain product excellence.

d. Tempering/thawing of frozen beef shall comply with the following:

-microwave tempering/thawing of frozen beef is permitted provided excellent condition of the beef, as determined by the USDA, is maintained
-the microwave tempering/thawing process shall not exceed 48 hours from time of removal of the beef from the freezer to further preparation
-regardless of the process used, frozen beef shall be tempered/thawed to a temperature not to exceed 40°F
-when a tempering/thawing process other than microwave is used, the time from the start of the tempering/thawing process (removal from freezer) to further preparation shall not exeed 96 hours
-beef which has been tempered/thawed shall not be refrozen.

* 3.2.2 Meatballs, cooked, bulk frozen. Meatballs received in a cooked, frozen state shall be accompanied by a USDA Agricultural Marketing Service (AMS) Certificate certifying the requirements of this paragraph and 3.2.1, 3.2.1.1, 3.3.1, 3.3.2, and 3.3.3 have been met. The cooked meatballs shall be packaged and heat sealed in USDA approved food packaging material within 4 hours after the cooking process. Within 24 hours after packaging, the packaged meatballs shall have been frozen to an internal temperature of 0°F or below. The cooked, packaged, frozen meatballs shall be held at an internal temperature of 0°F or lower for not more than 60 days prior to pouch filling.

3.2.2.1 Meatballs, cooked, individually quick frozen (IQF). IQF cooked meatballs shall comply with all the requirements of 3.2.2, except that the meatballs shall have been individually quick frozen and packaged in USDA approved food packaging material immediately after cooking.

3.2.3 Rice, medium grain. Rice shall be parboiled medium grain rice, U.S. No. 2 or better of the U.S. Standards for Milled Rice and of such varieties as Blue Rose, Calrose, Nato, Nova, Saturn, Brozzos, Vista, and other related

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varieties. The rice shall contain not more than 2 percent of the kernels having white ungelatinized areas and not less than 10 percent or more than 15 percent moisture.

3.2.4 Allspice, ground. Ground allspice shall be derived from dried, nearly ripe fruit of Pimenta officinalis Lindi and shall possess a fragrant, clove-like aroma, strongly aromatic, pungent, clove-like flavor, and a dark reddish-brown color. Allspice shall contain not less than 3.0 mL of volatile oil per 100 grams of ground allspice with 80.0 percent eugenol as a principal constituent. A minimum of 95 percent, by weight, shall pass through a U.S. Standard No. 25 sieve.

3.2.5 Cinnamon, ground. Ground cinnamon shall be prepared from the dried bark of Cinnamomum burmanii Blume, commonly known as "Korintz Cinnamon". The cinnamon shall contain not less than 1.5 mL of volatile oil per 100 grams of ground cinnamon and shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 60 sieve.

3.2.6 Clove, ground. Ground cloves shall be prepared from the dried, unopened flower buds of Caryophyllus aromatic L. The powder shall be dark reddish-brown in color and shall possess a strong aromatic odor with a hot pungent taste. The ground cloves shall contain not less than 15.0 mL of volatile oil per 100 grams of ground cloves and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 30 sieve.

3.2.7 Fat, frying. Frying fat shall be prepared from animal or vegetable (excluding coconut oil, palm oil, and palm kernel oil) fats, or a blend of animal and vegetable fats, which have been hydrogenated and deodorized. The frying fat shall have a clean bland flavor. The frying fat may be liquid or plasticized, may contain antioxidants in combinations and amounts allowed by FDA, and shall contain antifoaming agents. The frying fat shall have a maximum free fatty acid content of 0.05 percent (oleic), a stability of not less than 40 hours (A.O.M.), and a smoke point of not less than 400°F.

3.2.8 Flavoring, smoke. Smoke flavoring shall be an aqueous solution of natural maple and/or oak wood smoke and shall be dark brown-black in color with a clean smoke odor. The smoke flavoring shall have a titratable acidity of 6.2 to 7.2 percent expressed as acetic acid, a carbonyl level (butanone -2) of 7.0 to 8.0 grams per 100 mL, and a phenolic level (dimethoxyphenol) of 7.5 to 9.0 mg per mL.

3.2.9 Garlic powder. Garlic powder shall be Fancy Grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.10 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

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3.2.11 Mustard flour. Mustard flour shall be a bright yellow powder prepared from a blend derived from the endosperm of the seed of Brassica hirta Moench and Brassica juncea. Mustard flour shall contain not less than 0.4 mL volatile oil per 100 grams of mustard flour and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 60 sieve.

3.2.12 Oil, vegetable. Vegetable oil (excluding coconut oil, palm oil, and palm kernel oil) shall possess a clean, bland flavor and shall have a minimum stability of 15 hours A.O.M., a free fatty acid value not to exceed 0.05 percent, a moisture/volatile matter content not to exceed 0.06 percent by weight and pass the cold test of 5.5 hours (minimum).

3.2.13 Onion, dehydrated, chopped. Dehydrated chopped onion shall be Fancy Grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.14 Pepper, red, ground. Ground red pepper shall be derived from red, ripe fruit of Capsicum frutescens L. and shall possess the characteristic yellowish-red to red color with a perceptible sensation of pungency. The Scoville Pungency Value shall be not less than 30,000 units. The red pepper shall be uniformly ground to allow a minimum of 95 percent, by weight, to pass through a U.S. Standard No. 40 sieve and not less than 95 percent, by weight, to be retained on a U.S. Standard No. 60 sieve.

3.2.15 Salt. Salt shall be noniodized, white, refined sodium chloride, with or without anticaking agents and shall comply with purity standards for sodium chloride of the Food Chemicals Codex.

3.2.16 Sauce, Worcestershire. Worcestershire sauce shall be brown to dark brown liquid and shall possess a pleasant, tart, peppery, fruit-spice flavor with a typical heavy viscosity. The titratable acidity (as acetic acid) shall not be less than 2.8 nor more than 3.8 percent.

3.2.17 Starch, food, modified, high opacity. High opacity starch shall be white, odorless, finely pulverized, modified maize food starch for use in thermostabilized foods and shall comply with MIL-STD-900. The modified starch shall demonstrate initial viscosity development in the temperature range 140°F to 170°F and a typical viscosity (be fully hydrated) at common retort temperatures. The modified starch shall resist breakdown at low pH and under shear stress. The cooked modified starch slurry shall be bland with essentially no cereal or starch taste.

3.2.18 Sugar, brown, light. Brown sugar shall be partially refined cane or beet sugar. The sugar shall be light brown in color and shall possess a sweet, molasses-like flavor.

3.2.19 Tomato paste. The tomato paste shall be U.S. Grade A of the U.S. Standards for Grades of Tomato Paste and shall be of the latest season's pack. See formulation (3.3.4) for soluble solids content.

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3.2.20 Vinegar, cider. Cider vinegar shall be prepared from apple cider and be clear, light amber in color, and be free from haze, sediment, suspended matter, and floating particles. The cider vinegar shall contain not less than 4.0 percent and not more than 5.0 percent acetic acid.

3.2.21 Water. Water used for formulation, blanching, ice making, and washing shall conform to the National Primary Drinking Water Regulations.

3.2.22 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this specification. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

* 3.2.23 Starch, waxy maize, modified, filling and processing aid. Starch shall be white to off-white, odorless, finely pulverized powder suitable for use as a filling and processing aid in thermostabilized foods. When hydrated at temperatures of 165°F to 200°F, the starch shall develop a paste. The initial viscosity of the paste shall be retained under conditions of moderate shear and prolonged holding times. During retorting at temperatures in excess of 200°F, the starch shall undergo extensive to complete breakdown of the initial viscosity. The starch shall be bland, shall demonstrate easy dispersibility, and shall comply with MIL-STD-900.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Beef preparation. Boned and trimmed beef which meets the requirements of 3.2.1 shall be further trimmed, if necessary, so as to ensure compliance with finished product requirements. The boned and trimmed beef shall be mechanically ground through a plate having holes of 3/16 inch diameter. After grinding, the beef shall be maintained for not more than 24 hours at an internal temperature of 28°F to 40°F prior to preparation of the meatballs (see 3.3.3).

3.3.2 Rice preparation. The rice shall be prepared as follows:

a. Batch weigh the rice (dry).
b. To the weighed rice, add an equal weight of warm water.
c. Soak the rice for a period of approximately 1/2 hour (or less time if the rice has completely absorbed all of the water).
d. Add the total amount of soaked rice and unabsorbed water to the meatball formulation (see 3.3.3) as the combined percent by weight of rice.
e. The soaked rice and unabsorbed water shall be combined with the other meatball formulation ingredients within 2 hours after completion of soaking.

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3.3.3 Meatball preparation. The meatballs shall be formulated and prepared as follows:

Ingredient	Percent by weight
Beef, ground	78.00
Rice, hydrated	18.95
Onions, chopped, dehydrated	1.80
Salt	1.25

a. The ingredients shall be combined and thoroughly mixed. The mixture shall be formed into meatballs approximately 5/8 inch in diameter. If the meatballs are not immediately formed and cooked after mixture preparation, the internal temperature of the mixture and formed meatballs shall be rapidly lowered to 28°F to 40°F and maintained at this internal temperature range for not more than 12 hours prior to the browning (cooking) process.

b. The meatballs shall be surface browned until no pink color is visible in the interior of the meatball. The cooked yield of the meatballs shall be approximately 85 percent. The meatballs may be cooked by any commercial method except moist heat (steaming). Frying fat, if used, shall comply with requirements of 3.2.7. The fat shall be filtered at least daily and the frying system cleaned at least weekly. The fat shall be discarded when foaming, undesirable odors and flavors or other evidence of fat breakdown is apparent.

c. If a hot fill system is used, the cooked meatballs shall be held in the temperature range of 150°F to 180°F for a period of time not to exceed 2 hours.

d. If a cold fill system is used, the cooked meatballs shall immediately be cooled to an internal temperature of 28°F to 40°F and held at this temperature for not more than 48 hours prior to product preparation.

3.3.4 Spicy tomato sauce preparation. The sauce shall be formulated and prepared as follows:

Ingredients	Percent by weight
Tomato paste (24% solids) 1/	36.00
Water	33.70
Sugar, brown, light	11.00
Vinegar, cider	8.30
Oil, vegetable	3.00
Starch, modified	2.00
Salt 2/	1.50

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Sauce, Worcestershire	1.50
Onion, dehydrated, chopped	0.95
Flavoring, smoke	0.50
Mustard flour	0.50
Garlic powder	0.50
Lecithin	0.30
Pepper, red, ground	0.10
Clove, ground	0.10
Allspice, ground	0.03
Cinnamon, ground	0.02

1/ If the specific tomato percent soluble solids are not available, the tomato solids that are used must be calculated to bring the solids level to that specified.

2/ The total amount of salt in the sauce formula shall be adjusted as necessary to produce a product that complies with the finished product salt requirements (see 3.6).

NOTE: The following sauce preparation procedures were used in the development of this product. Alternate procedures may be used provided finished product requirements are met. (When alternate procedures are used, the time and temperature requirements specified for the prepared sauce are still applicable.)

a. The starch and part of the water shall be mixed to form a slurry.

b. The balance of the water shall be heated in a steam-jacketed kettle until the temperature reaches 180°F to 190°F.

c. The starch slurry shall be added to the water and the mixture stirred continuously until thickened and the temperature reaches 180°F to 190°F.

d. The steam supply to the kettle shall be shut off and the balance of ingredients added and uniformly mixed to form the sauce.

e. The volume of the sauce shall be adjusted with water prior to filling to compensate for evaporation loss during heating and holding.

f. The sauce shall be filled within 4 hours after cooking. If a hot fill method is used, the prepared sauce shall be held at a temperature of not less than 150°F nor greater than 180°F at all times prior to filling into the pouches. If a cold fill method is used, the prepared sauce shall be immediately cooled to 28°F to 40°F and held in this temperature range at all times prior to filling into the pouches.

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* 3.3.4.1 Filling and processing aid starch preparation. When a filling and processing aid starch is used, a solution (slurry) shall be prepared with the filling and processing aid starch and part of the sauce formulation water in a concentration adequate to hold ingredients in suspension for uniform distribution during mixing and pumping of product.

3.4 Pouch filling and sealing. Each pouch (see 5.1) shall be filled with product such as to conform to the finished product requirements and to the following requirements:

a. Each pouch shall be filled and sealed in accordance with the pouch filling and sealing requirements for class 1 as specified in MIL-P-44073.

b. A hot fill system is permitted if it is compatible with the single-stage or multi-stage filling and sealing method used.

c. The temperature of the meatballs and spicy tomato sauce at the time of filling shall be as specified (see 3.3.3 and 3.3.4) for the applicable filling process used. When frozen cooked meatballs are used, they shall be thawed/tempered only to the extent necessary to accommodate the filling operation. Microwave tempering/thawing is permitted provided that requirements specified in 3.2.1.1d are met. Blanching is not permitted.

d. The product may be filled into the pouch using either a single-stage or multi-stage filling operation.

e. Each filled and sealed pouch shall be in the retort process within 2 hours after sealing.

3.5 Pouch thermoprocessing. The filled and sealed pouches shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6 has been achieved. The thermoprocessing operation shall be in compliance with the thermoprocessing requirements for class 1 of MIL-P-44073.

3.6 Finished product requirements. The finished product shall comply with the following requirements.

a. There shall be no foreign material, such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor, such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, in any individual pouch shall be not more than 0.35 ounce.

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e. No individual pouch shall contain a bone piece measuring 0.3 inch or more in any dimension.

f. The average net weight shall be not less than 8.0 ounces.

g. No individual pouch shall contain less than 7.5 ounces of product.

h. No individual pouch shall contain less than 3.5 ounces of drained meatballs.

i. The average drained weight of meatballs shall be not less than 4.2 ounces.

j. At least 75 percent, by weight, of the meatballs shall be intact.

k. Meatballs shall be uniform in size and shape.

l. The texture of the meatballs shall not be dry, mushy, rubbery, or tough.

m. The consistency of the spicy tomato sauce shall be smooth, without lumps, and shall not be excessively thick or thin.

n. The average fat content of the finished product shall be not greater than 10.0 percent.

o. The fat content of the product in any individual pouch shall be not greater than 12.0 percent.

p. The salt content of the product in any individual pouch shall be not greater than 1.8 percent nor less than 1.2 percent.

q. The product shall show no evidence of excessive heating (materially darkened or scorched).

3.6.1 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The beef component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under the Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

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* 3.3.4.1 Filling and processing aid starch preparation. When a filling and processing aid starch is used, a solution (slurry) shall be prepared with the filling and processing aid starch and part of the sauce formulation water in a concentration adequate to hold ingredients in suspension for uniform distribution during mixing and pumping of product.

3.4 Pouch filling and sealing. Each pouch (see 5.1) shall be filled with product such as to conform to the finished product requirements and to the following requirements:

a. Each pouch shall be filled and sealed in accordance with the pouch filling and sealing requirements for class 1 as specified in MIL-P-44073.

b. A hot fill system is permitted if it is compatible with the single-stage or multi-stage filling and sealing method used.

c. The temperature of the meatballs and spicy tomato sauce at the time of filling shall be as specified (see 3.3.3 and 3.3.4) for the applicable filling process used. When frozen cooked meatballs are used, they shall be thawed/tempered only to the extent necessary to accommodate the filling operation. Microwave tempering/thawing is permitted provided that requirements specified in 3.2.1.1d are met. Blanching is not permitted.

d. The product may be filled into the pouch using either a single-stage or multi-stage filling operation.

e. Each filled and sealed pouch shall be in the retort process within 2 hours after sealing.

3.5 Pouch thermoprocessing. The filled and sealed pouches shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6 has been achieved. The thermoprocessing operation shall be in compliance with the thermoprocessing requirements for class 1 of MIL-P-44073.

3.6 Finished product requirements. The finished product shall comply with the following requirements.

a. There shall be no foreign material, such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor, such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, in any individual pouch shall be not more than 0.35 ounce.

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e. No individual pouch shall contain a bone piece measuring 0.3 inch or more in any dimension.

f. The average net weight shall be not less than 8.0 ounces.

g. No individual pouch shall contain less than 7.5 ounces of product.

h. No individual pouch shall contain less than 3.5 ounces of drained meatballs.

i. The average drained weight of meatballs shall be not less than 4.2 ounces.

j. At least 75 percent, by weight, of the meatballs shall be intact.

k. Meatballs shall be uniform in size and shape.

l. The texture of the meatballs shall not be dry, mushy, rubbery, or tough.

m. The consistency of the spicy tomato sauce shall be smooth, without lumps, and shall not be excessively thick or thin.

n. The average fat content of the finished product shall be not greater than 10.0 percent.

o. The fat content of the product in any individual pouch shall be not greater than 12.0 percent.

p. The salt content of the product in any individual pouch shall be not greater than 1.8 percent nor less than 1.2 percent.

q. The product shall show no evidence of excessive heating (materially darkened or scorched).

3.6.1 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The beef component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under the Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of AOAC International and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

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4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Filled and sealed pouch inspection. Inspection of filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-44073.

4.5.4 Net weight inspection. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 7.5 ounces shall be scored as a minor defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 0.1 ounce. In addition, the lot shall be rejected if the sample average net weight is less than 8.0 ounces.

4.5.5 Product and drained weight inspection.

4.5.5.1 Product inspection. The filled and sealed sample pouches shall be held for a minimum of 72 hours at room temperature ($65^{\circ}F$ to $75^{\circ}F$) after completion of the thermoprocessing operation. The pouches shall be heated in $180^{\circ}F$ to $190^{\circ}F$ water for 10 minutes, opened, and inspected. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The sample size shall be as indicated by the sampling plan specified in table III. The contents shall be inspected for defects indicated in table IV.

TABLE III. Double sampling plan for product inspection 1/

Lot size (pouches)	Sample size (pouches)	Cumulative sample	Acceptance number	Rejection number
0 to 500,000	20	--	0	2
	20	40	1	2

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples

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shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE IV. Product defects 1/ 2/

Category	Defect
Major	
101	Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, in a pouch is more than 0.35 ounce.
102	Presence of a bone piece measuring 0.3 inch or more in any dimension.
103	Less than 75 percent, by weight, of the meatballs are intact.
104	Meatballs are not uniform in appearance (size and shape).
105	Meatball texture is dry, tough, rubbery, or mushy.
106	Sauce consistency is lumpy or is excessively thick or thin.
107	Product shows evidence of excessive heating (materially darkened or scorched).

1/ The presence of foreign material (for example, glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot (see 3.6.1).

* 4.5.5.2 Drained weight inspection. The filled and sealed sample pouches shall be held for a minimum of 72 hours at room temperature (65°F to 75°F) after completion of the thermoprocessing operation. The pouches shall be heated in 180°F to 190°F water for approximately 10 minutes, opened, and inspected. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The sample size shall be as indicated by the double sampling plan specified in table V. The drained weight shall be determined and inspected for the defect listed in table VI.

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TABLE V. Double sampling plan for drained weight inspection 1/

Lot size (pouches)	Sample size (pouches)	Cumulative sample	Acceptance number	Rejection number
0 to 500,000	20	--	0	2
	20	40	1	2

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE VI. Drained weight defect 1/

Category	Defect
Major	
101	Drained weight of meatballs in a pouch is less than 3.5 ounces. 2/

1/ If the sample average drained weight of meatballs is less than 4.2 ounces, the lot shall be rejected.

2/ To determine drained weight, the pouch contents shall be poured into a flat bottom container. A minimum of three times the pouch's volume of 180° to 190°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated so as to liquify rendered fat and to remove the sauce. The contents shall then be poured into a U.S. Standard No. 8 sieve in a manner that will distribute the product over the sieve. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at an approximate 45 degree angle and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight shall be reported to the nearest 0.1 ounce.

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* 4.5.6 Fat and salt content testing. Eight randomly selected, unopened pouches shall be warmed in a water bath to melt fat adhering to the inside of the pouches. The sample pouches shall be opened and the entire contents of each pouch shall be separately blended in a Waring blendor or equivalent. The fat and salt content of the product from each pouch in the sample shall be determined in accordance with the Official Methods of Analysis of AOAC International, method numbers 960.39 and 935.47. Test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the fat or salt requirement in 3.6 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.7 Packaging inspection. Inspection of packaging shall be in accordance with the quality assurance provisions of MIL-P-44073.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the requirements of MIL-P-44073.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the soliciation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.6.1 and 6.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Subject term (key word) listing.

Combat field feeding
Food processing
Operational rations
Pouch

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6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, QM
Navy - MC
DP - SS

Preparing activity:

Army - GL

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	MIL-M-44203B	1992 July 23

3. DOCUMENT TITLE MEATBALLS, BEEF AND RICE, IN SPICY TOMATO SAUCE, SMOKE FLAVORING ADDED, THERMOSTABILIZED, FOR MEAL, READY-TO-EAT

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

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8. PREPARING ACTIVITY

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c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WTP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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